# Create Load Balance in ApiGateWay

## 1- Original config

```
server.port=8082

app.resourceserver.base-url=http://localhost:8081
app.photoserver.base-url=http://localhost:8090
app.albumserver.base-url=http://localhost:8091

# ===== WebMVC Gateway routes (NOT spring.cloud.gateway.routes) =====

spring.cloud.gateway.server.webmvc.routes[0].id=users-status-check
spring.cloud.gateway.server.webmvc.routes[0].uri=${app.resourceserver.base-url}
spring.cloud.gateway.server.webmvc.routes[0].predicates[0]=Path=/users/**
spring.cloud.gateway.server.webmvc.routes[0].predicates[1]=Method=GET
spring.cloud.gateway.server.webmvc.routes[0].filters[0]=RemoveRequestHeader=Cookie

spring.cloud.gateway.server.webmvc.routes[1].id=photos
spring.cloud.gateway.server.webmvc.routes[1].uri=${app.photoserver.base-url}
spring.cloud.gateway.server.webmvc.routes[1].predicates[0]=Path=/photos/**
spring.cloud.gateway.server.webmvc.routes[1].predicates[1]=Method=GET
spring.cloud.gateway.server.webmvc.routes[1].filters[0]=RemoveRequestHeader=Cookie

spring.cloud.gateway.server.webmvc.routes[2].id=albums
spring.cloud.gateway.server.webmvc.routes[2].uri=${app.albumserver.base-url}
spring.cloud.gateway.server.webmvc.routes[2].predicates[0]=Path=/albums/**
spring.cloud.gateway.server.webmvc.routes[2].predicates[1]=Method=GET
spring.cloud.gateway.server.webmvc.routes[2].filters[0]=RemoveRequestHeader=Cookie

spring.application.name=ApiGateWay
eureka.client.serviceUrl.defaultZone: http://localhost:8010/eureka
```

## 2- Update load balance : uri=lb://ResourceServer ចឹងក្នុង Spring cloud វាដឹងថាជា lb: load balance ហើយ

```
server.port=8082

spring.cloud.gateway.server.webmvc.routes[0].id=users-status-check
spring.cloud.gateway.server.webmvc.routes[0].uri=lb://ResourceServer
spring.cloud.gateway.server.webmvc.routes[0].predicates[0]=Path=/users/**
spring.cloud.gateway.server.webmvc.routes[0].predicates[1]=Method=GET
spring.cloud.gateway.server.webmvc.routes[0].filters[0]=RemoveRequestHeader=Cookie

spring.cloud.gateway.server.webmvc.routes[1].id=photos
spring.cloud.gateway.server.webmvc.routes[1].uri=lb://Photos
spring.cloud.gateway.server.webmvc.routes[1].predicates[0]=Path=/photos/**
spring.cloud.gateway.server.webmvc.routes[1].predicates[1]=Method=GET
spring.cloud.gateway.server.webmvc.routes[1].filters[0]=RemoveRequestHeader=Cookie

spring.cloud.gateway.server.webmvc.routes[2].id=albums
spring.cloud.gateway.server.webmvc.routes[2].uri=lb://Albums
spring.cloud.gateway.server.webmvc.routes[2].predicates[0]=Path=/albums/**
spring.cloud.gateway.server.webmvc.routes[2].predicates[1]=Method=GET
spring.cloud.gateway.server.webmvc.routes[2].filters[0]=RemoveRequestHeader=Cookie

spring.application.name=ApiGateWay
eureka.client.serviceUrl.defaultZone: http://localhost:8010/eureka
```

Testing Postman:

OAuth-ApiGateWay > 1-Resource

GET http://localhost:8082/users/status/check

Docs Params Authorization Headers 8 Body Scripts Settings

Auth Type

Bearer Token

Token

The authorization header will be automatically generated when you send the request. Learn more about Bearer Token authorization.

Body Cookies Headers 5 Test Results 200 OK 18 ms 185 B

Raw Preview Visualize

```
1  Working on port: 1292
```

GET http://localhost:8082/users/status/check

Docs Params Authorization Headers 8 Body Scripts Settings

Auth Type

Bearer Token

Token

The authorization header will be automatically generated when you send the request. Learn more about Bearer Token authorization.

Body Cookies Headers 5 Test Results 200 OK 10 ms 185 B

Raw Preview Visualize

```
1  Working on port: 4463
```

ចឹង ApiGateWay វាចេះធ្វើការហើយដោយ switch service ម្តងម្នាក់

## DS Replicas

localhost

## Instances currently registered with Eureka

| Application | AMIs | Availability Zones | Status |
|---|---|---|---|
| APIGATEWAY | n/a (1) | (1) | UP (1) - 10.172.8.187:ApiGateWay:8082 |
| RESOURCESERVER | n/a (3) | (3) | UP (3) - ResourceServer:69c169d61c6ae65977e6b71ceea28939 , ResourceServer:3af002b853d72c28bf6737e12e667c6d , ResourceServer:936c616f2fe196d17f463474b953d9f0 |

# 3. Inject the Environment env មកប្រើ

```java
package com.vibol.api.ResourceServer.controllers;

import com.vibol.api.ResourceServer.response.UserRest;

@RestController
@RequestMapping("/users")
@Slf4j
public class UsersController {

    @Autowired
    Environment env;

    @GetMapping("/status/check")
    public String status(){
        log.info("Request From scope profile");
        return "Working on port: " + env.getProperty("local.server.port");
    }
```

In spring boot we can inject this Environment មកប្រើបាន

```java
package com.vibol.api.ResourceServer.controllers;

import com.vibol.api.ResourceServer.response.UserRest;
import lombok.extern.slf4j.Slf4j;

import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.core.env.Environment;
import org.springframework.security.access.annotation.Secured;
import org.springframework.security.access.prepost.PostAuthorize;
import org.springframework.security.access.prepost.PreAuthorize;
import org.springframework.security.core.annotation.AuthenticationPrincipal;
import org.springframework.security.oauth2.jwt.Jwt;
import org.springframework.web.bind.annotation.*;

@RestController
@RequestMapping("/users")
@Slf4j
public class UsersController {

        @Autowired
        Environment env;

    @GetMapping("/status/check")
    public String status(){
        log.info("Request From scope profile");
        return "Working on port: " + env.getProperty("local.server.port");
    }

//    @PreAuthorize("hasAuthority('ROLE_developer') or #id == #jwt.subject")
    @PreAuthorize("#id == #jwt.subject")
//    @PreAuthorize("hasRole('developer')")
//    @Secured("ROLE_developer")
    @DeleteMapping(path="/{id}")
    public String deletedUser(@PathVariable String id, @AuthenticationPrincipal Jwt jwt){
        return "Deleted user with id: " + id + " and JWT subject " + jwt.getSubject();
    }

    @PostAuthorize("returnObject.userId == #jwt.subject")
    @GetMapping(path="/{id}")
    public UserRest getUser(@PathVariable String id, @AuthenticationPrincipal Jwt jwt){
        return new UserRest("Vibol","long","80373687-a5e1-43d6-bbd8-36d421e760c");
    }

}
```